Agent Follow-Up - Software Program Plan

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Project Directory: `/home/tish/projects/agent-follow-up`

1. Purpose

Define a realistic, automatable system that ensures OpenClaw agents provide proactive follow-up updates for delegated or background work, instead of depending on memory or manual operator prompting.

2. Problem Statement

Current behavior depends on the active agent remembering to check back in. That is not reliable enough to support trust. We need a machine-enforced mechanism that tracks background work and sends follow-ups on completion, failure, or overdue intervals.

3. Program Goals

- Guarantee follow-up behavior through automation rather than agent intention.
- Integrate cleanly with OpenClaw sessions, spawned background work, and local processes.
- Support milestone, blocker, overdue, and completion notifications.
- Preserve a provable audit trail of tracked tasks and emitted updates.

4. Deliverables for Phase 1

- Task registry format and storage model.
- Background-check watchdog script/service design.
- Notification decision rules.
- Verification and validation strategy.
- Initial SLC documentation set (this package).

5. Scope

In Scope

- Tracking delegated/background tasks started by an OpenClaw agent.
- Polling active task state on a scheduler.
- Sending proactive updates when tasks are overdue, blocked, completed, or failed.
- OpenClaw-friendly local implementation on this VM.

Out of Scope (Phase 1)

- Perfect semantic progress extraction from all tools.
- Cross-machine orchestration.
- Guaranteed recovery from every external messaging outage.
- Complex multi-tenant scheduling.

6. Implementation Strategy

1. Create a durable task registry (JSON file) under the OpenClaw workspace.
2. Create a watchdog/checker that reads the registry, inspects tracked tasks, and decides whether to notify.
3. Run the checker on a regular cadence using a scheduler (prefer cron or systemd timer).
4. Record follow-up events and task resolution back into durable state.
5. Later: expose tracked tasks in the Ops Dashboard.

7. Key Planning Decision

Recommended technical direction: use a scheduler-backed watchdog with a durable task registry. Do not rely on heartbeat alone and do not rely on agent memory.

8. Success Criteria

- When background work is registered, the system issues an update without Tish asking.
- Completion and failure updates are generated automatically when observable.

- Overdue tasks generate periodic status reminders at the configured cadence.
- The system behavior is provable through logs, state files, and real notifications.